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by [Ali Alzahmi](#) - Tuesday, 26 August 2025, 7:12 PM

The spread of multi-agent systems (MAS) in artificial intelligence has reflected the usefulness of organised communications. Not only the agents need to be able to share data but also intentions, goals and commitments. In order to meet this requirement, ACLs, in particular, the Knowledge Query and Manipulation Language (KQML) were invented, the protocol of which is based on the speech act theory (Kim et al., 2024). Unlike the case with regular programming calls, ACLs are concerned with the semantics of interaction and not with its syntax. The optimal merit of ACLs is that it confines a heterogeneous environment (Wang et al., 2025). The agents that have been programmed in different programming languages and which are designed to run on different systems will be capable of communicating with each other, provided they work with the same type of communication standard. In addition, ACLs can include semantically rich messages, as messages are performatives (except inform, request), e.g. converse or play-game, and the intention of the agent is made evident. Finally, systems that incorporate the use of ACLs scale better, since effectively the communication patterns do not have to undergo vast reengineering as more agents are added to the system. Along with their benefits, ACLs are disadvantaged in practice as well. They create the necessity of doing significant reasoning to the agents, rising the complexity of computer calculations. There can also be a given efficiency restriction: ACLs are also overhead relative to lightweight calls and can thus not be applied in real-time or performance-based applications (Younas et al., 2025). These drawbacks explain why ACLs remain most common in academic use cases, when compared to commercial use cases, where less complex mechanisms are more popular, such as an API or message queues. In Python or Java, calling methods are a simple, high-performance and predictable way of communicating between software components (Dong et al., 2025). They best suited suited in tightly coupled systems. The preference of ACLs, in their turn, is in dynamic open environments, where agents are subject to autonomous action that necessitates the interpretation of the goals of other agents. This is due to the fact that, though ACLs, such as KQML are not heavily used in the industry, they are also a major research contribution.

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